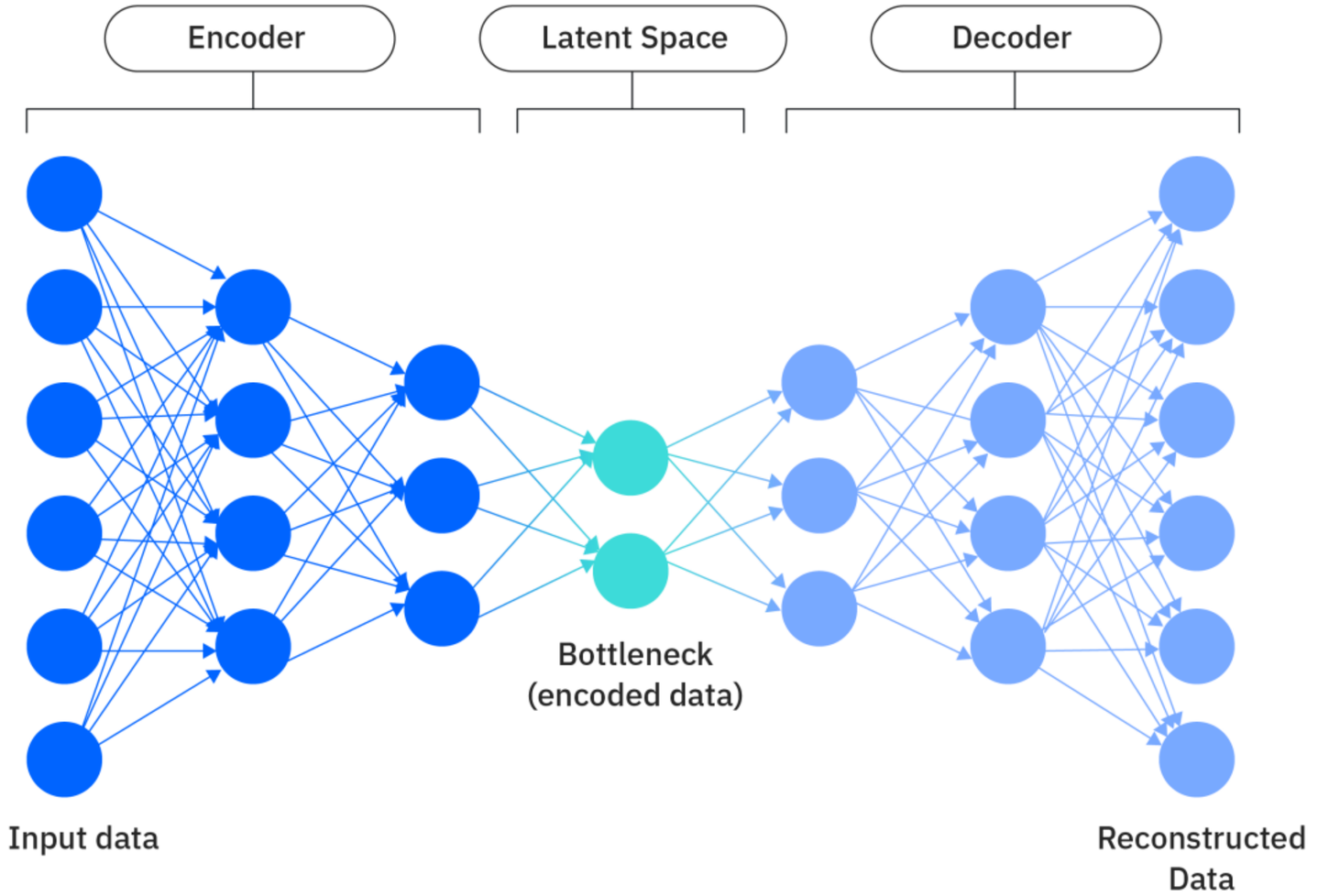
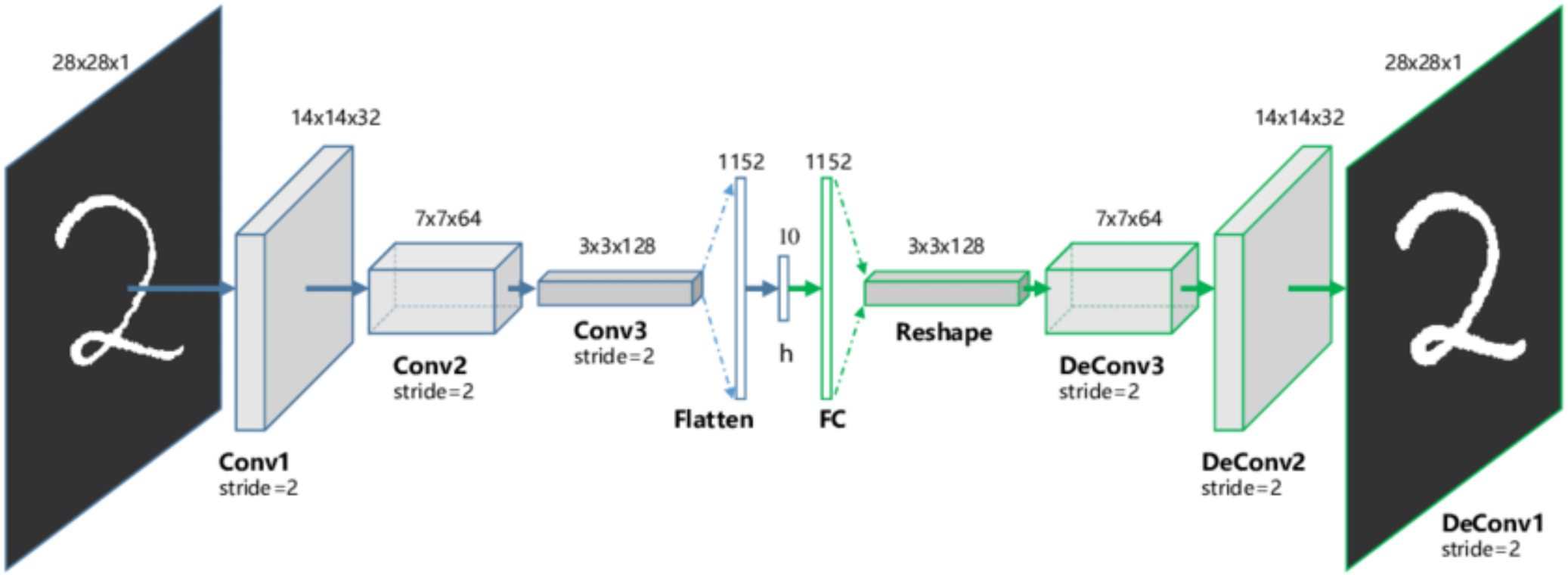
Audio Autoencoder

Autoencoder Architecture



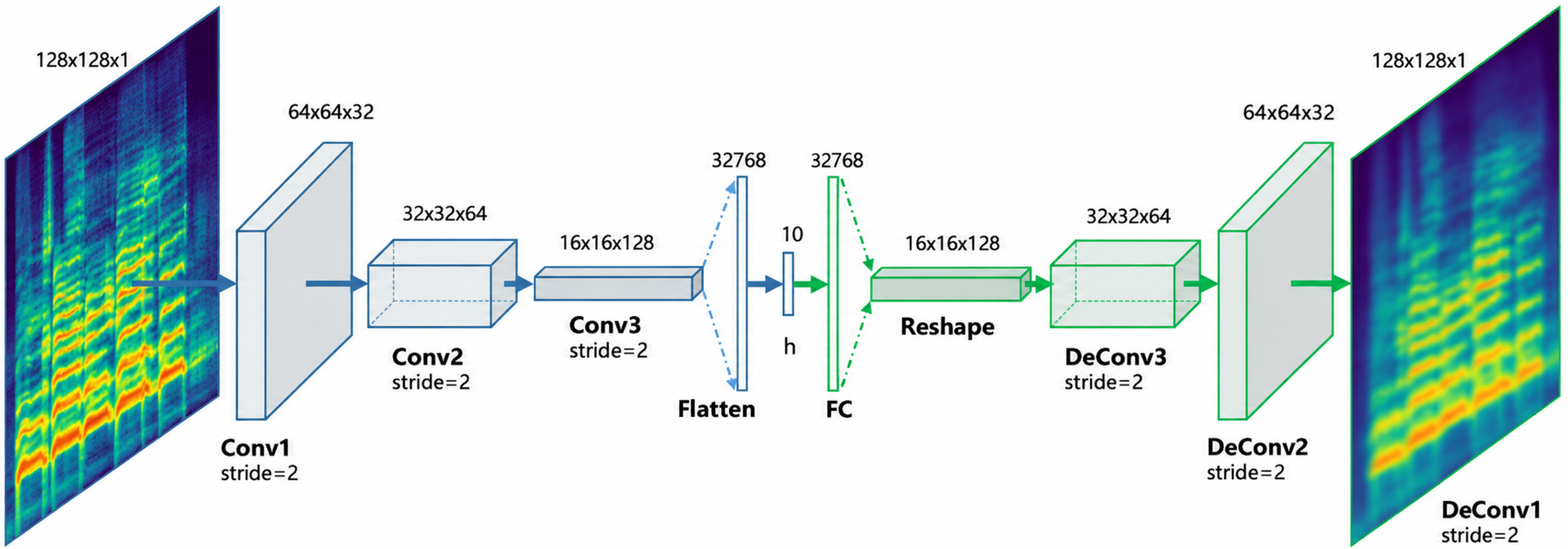
From: https://www.ibm.com/think/topics/latent-space

Convolutional Autoencoder



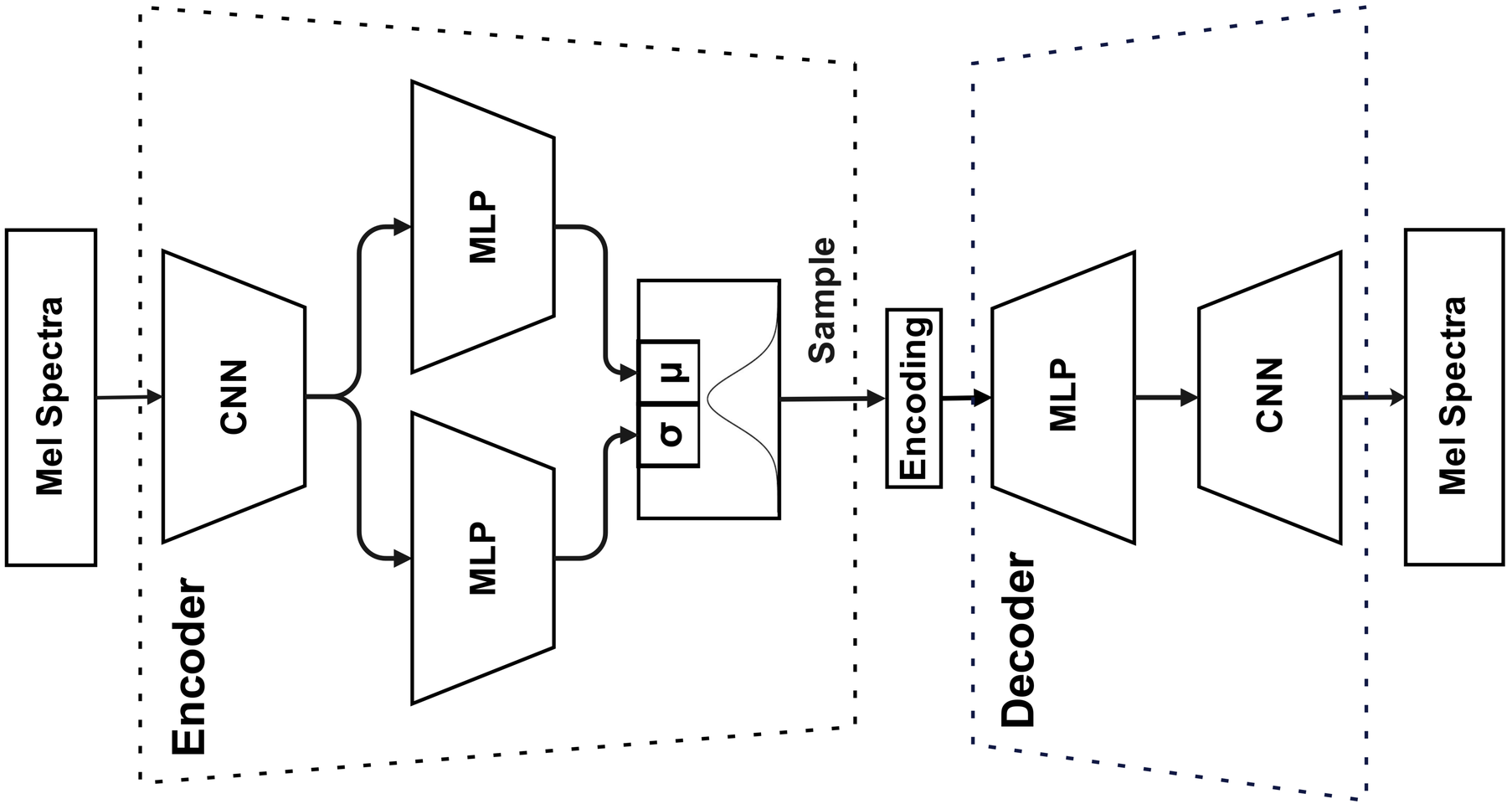
Convolution and Deconvolution Layers in an Autoencoder. ©Vaibhav Kumar

Convolutional Autoencoder

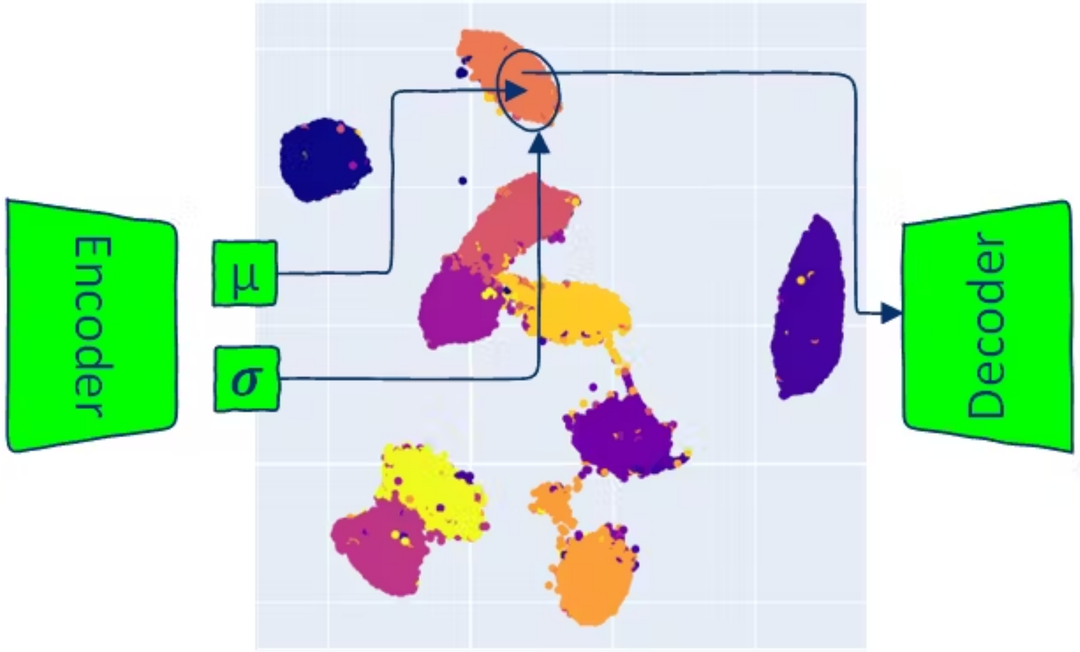


Convolution and Deconvolution Layers in an Autoencoder operating on Mel Spectra

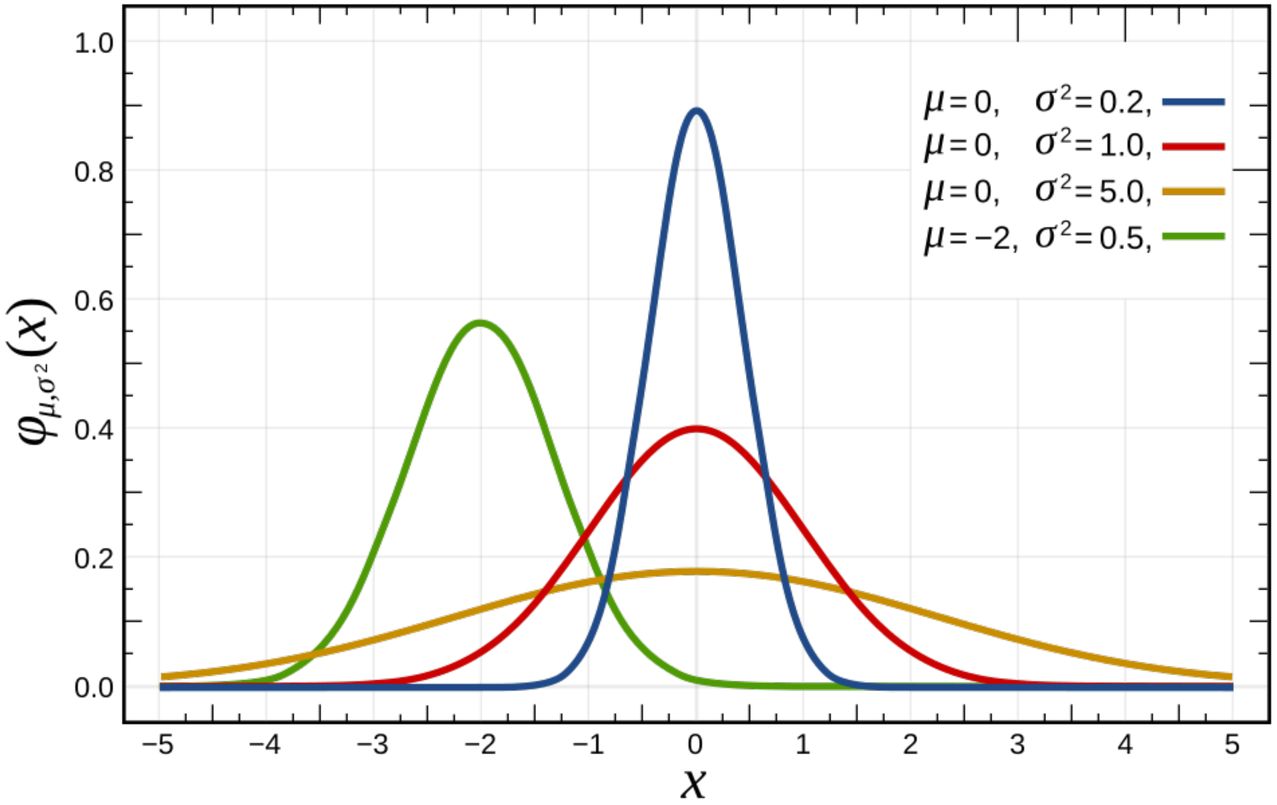
Audio Autoencoder



Variational Autoencoder



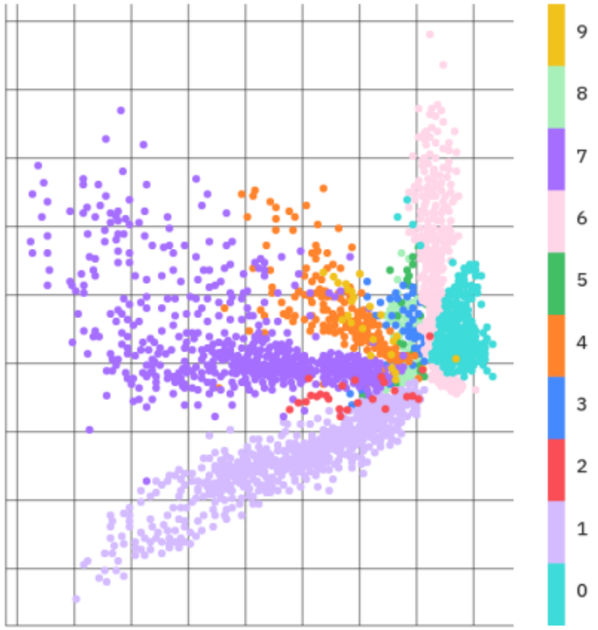
From https://hackernoon.com/how-to-sample-from-latent-space-with-variational-autoencoder



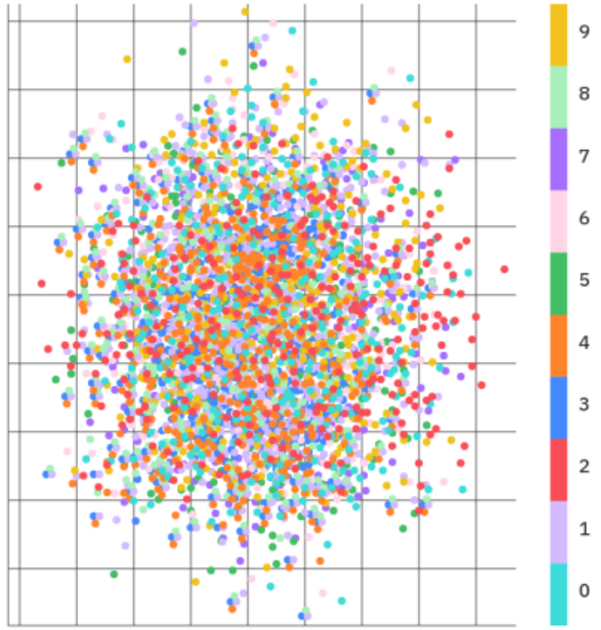
From https://en.wikipedia.org/wiki/Normal_distribution

Latent Space Distribution

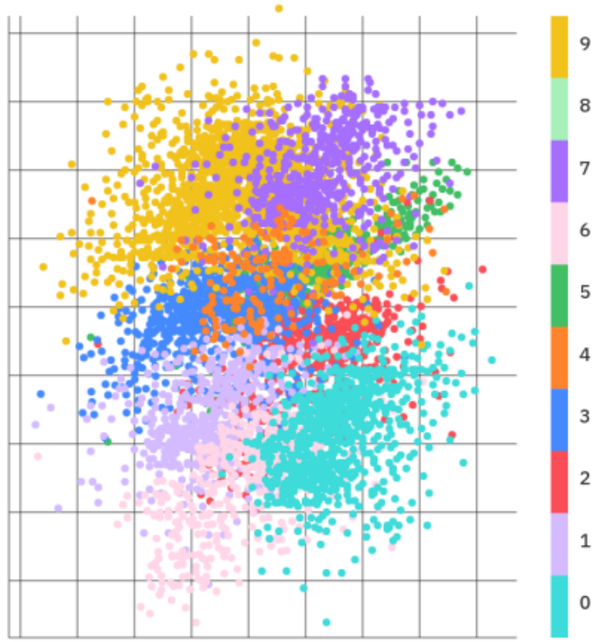
Only reconstruction loss



Only KL divergence

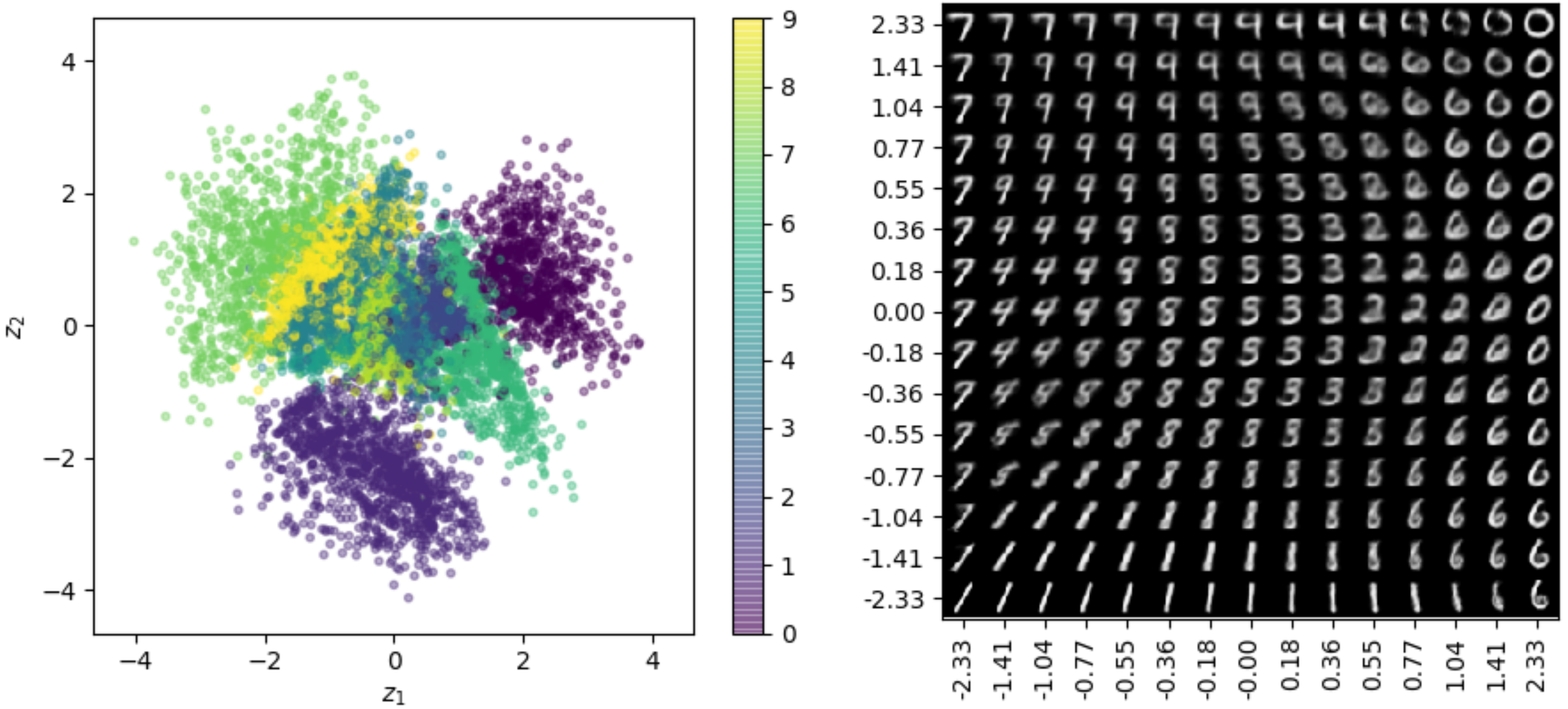


Reconstruction loss and KL divergence



From https://www.ibm.com/think/topics/latent-space

Latent Space



From: https://tiao.io/post/tutorial-on-variational-autoencoders-with-a-concise-keras-implementation/

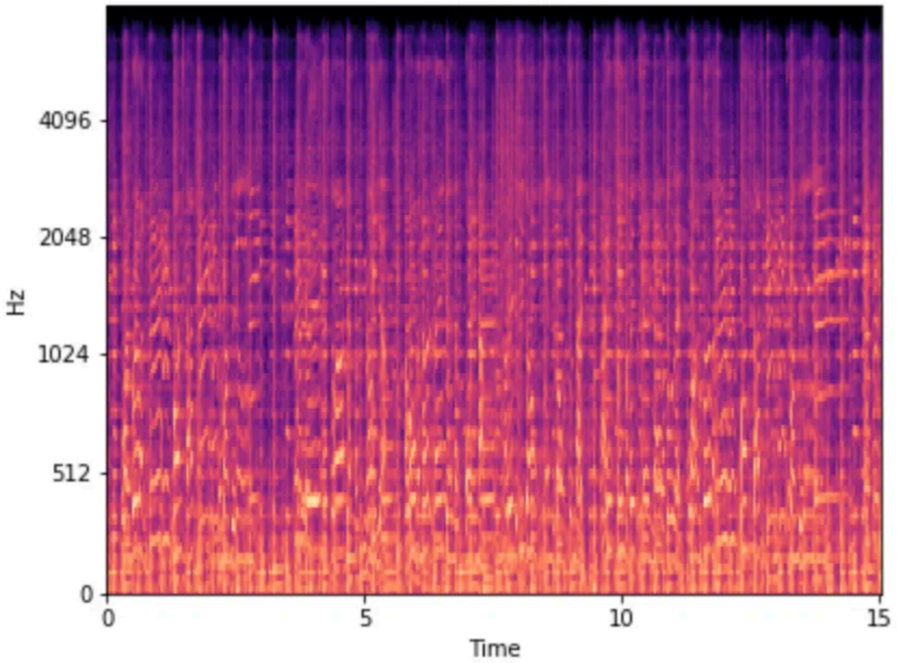
Latent Space



Bilinear Interpolation of Encodings of Images of Faces. ©Mathijs Pieters and Marco A. Wiering, 2018

Audio Vocoder





?

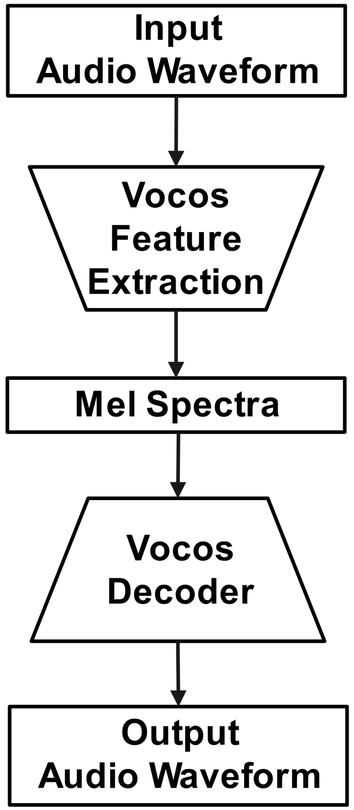


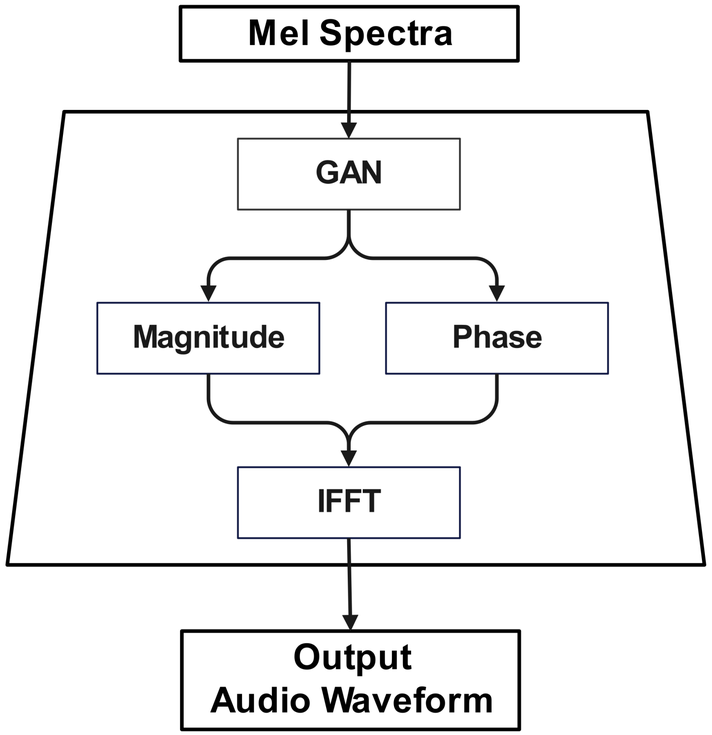
Audio Waveform

Mel Spectrogram

Audio Waveform

Vocos


Input
Audio Waveform
Vocos
Feature
Extraction
Mel Spectra
Vocos
Decoder
Output
Audio Waveform


Mel Spectra
GAN
Magnitude
Phase
IFFT
Output
Audio Waveform


kittn
/vocos-mel-48khz-alpha1
huggingface.co

Audio Vocoder

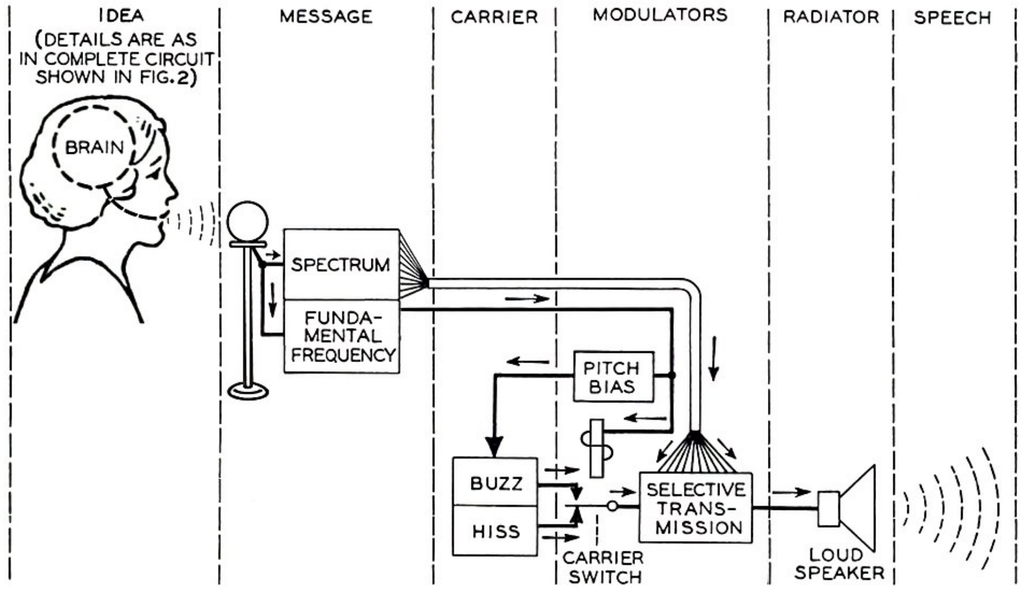
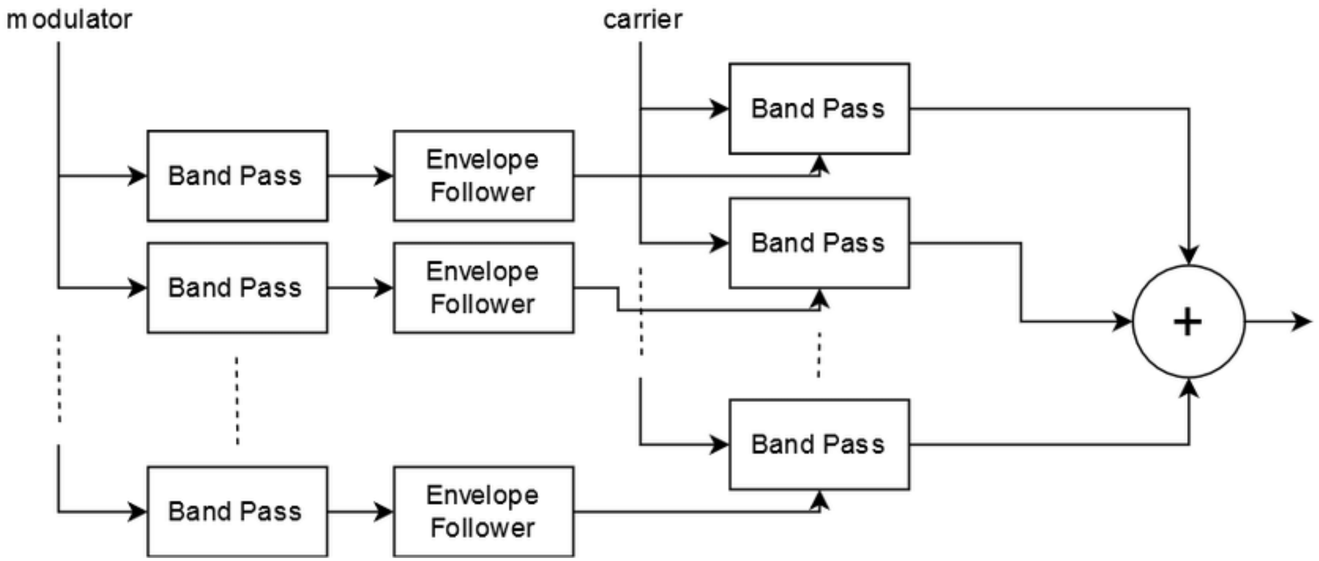


Fig. 7—Schematic circuit of the vocoder.

Dudley's Vocoder, based on: Dudley 1940



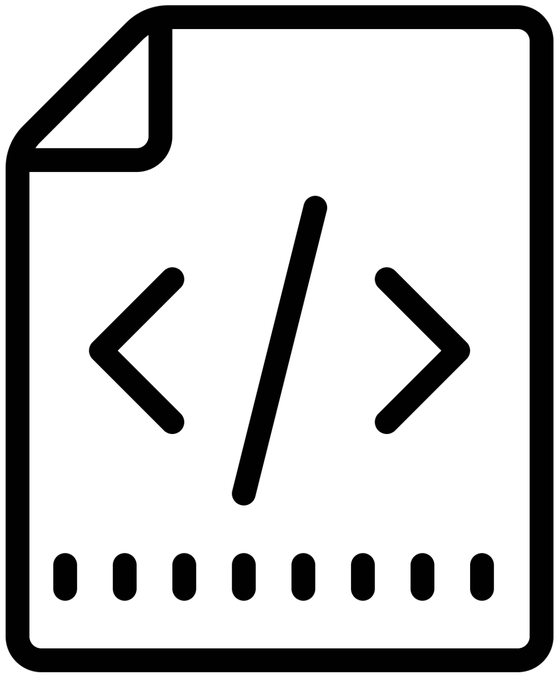
Schematic of a standard vocoder (Wikipedia)

Kraftwerk - Wir sind die Roboter (1978)

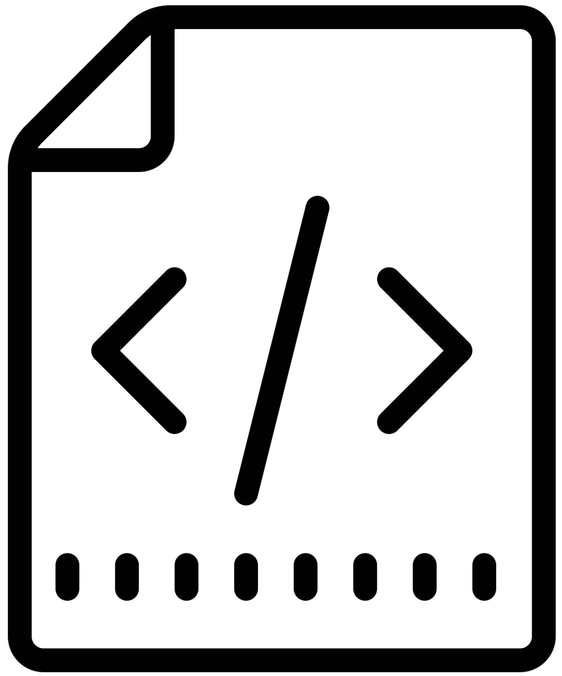


https://www.youtube.com/watch?v=JgtFRj38fEw

Audio Autoencoder - Demo - Training and Inference

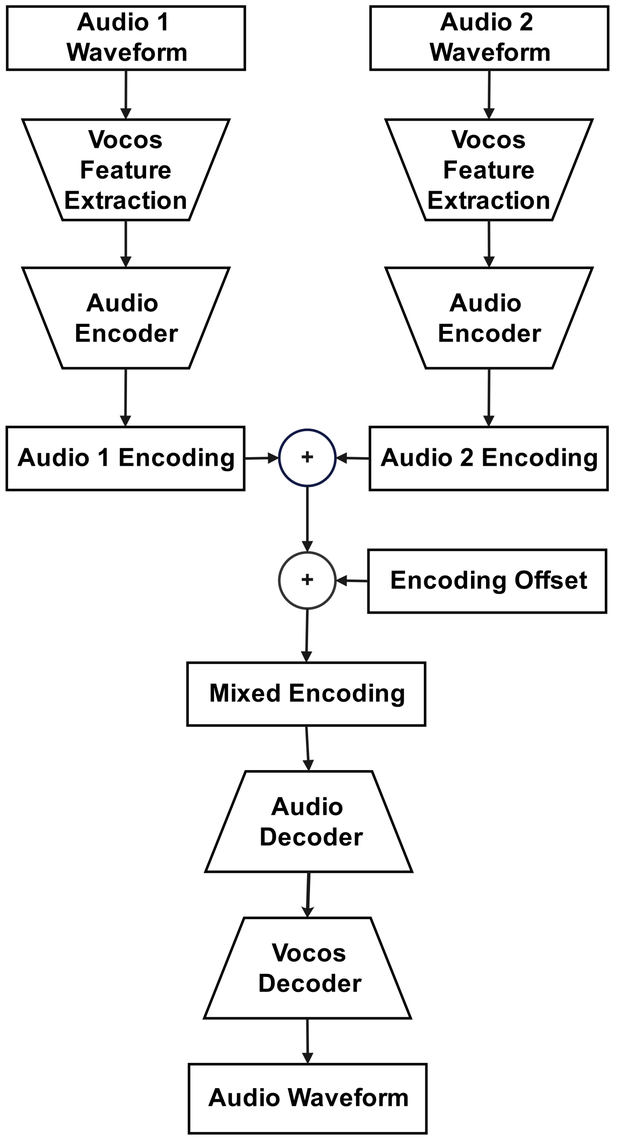


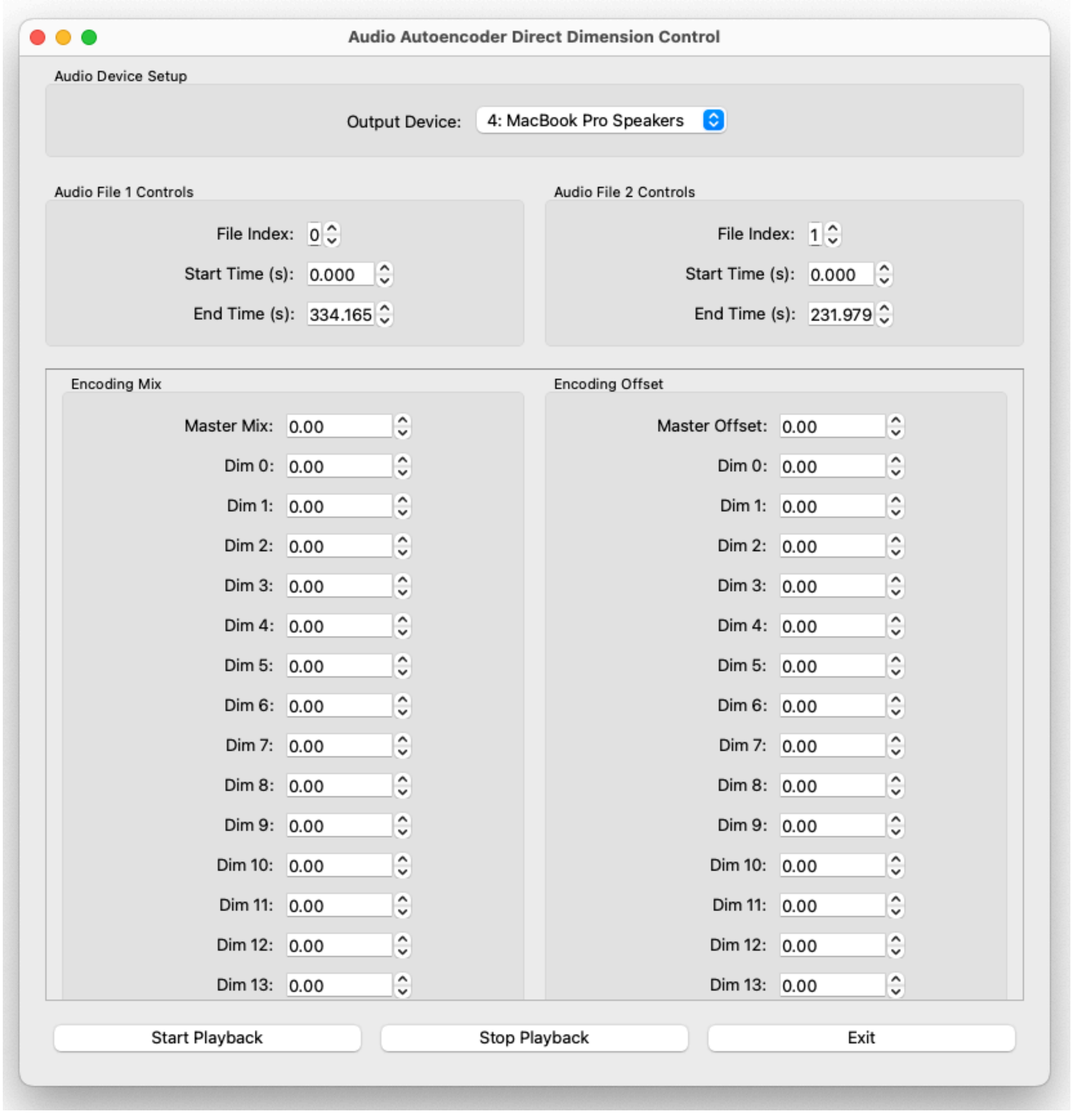
AI_Sound / Autoencoder / Training
audio_vae_vocos_cnn.py



AI_Sound / Autoencoder / Inference / HelloWorld
vae_vocos_cnn_hello_world.py

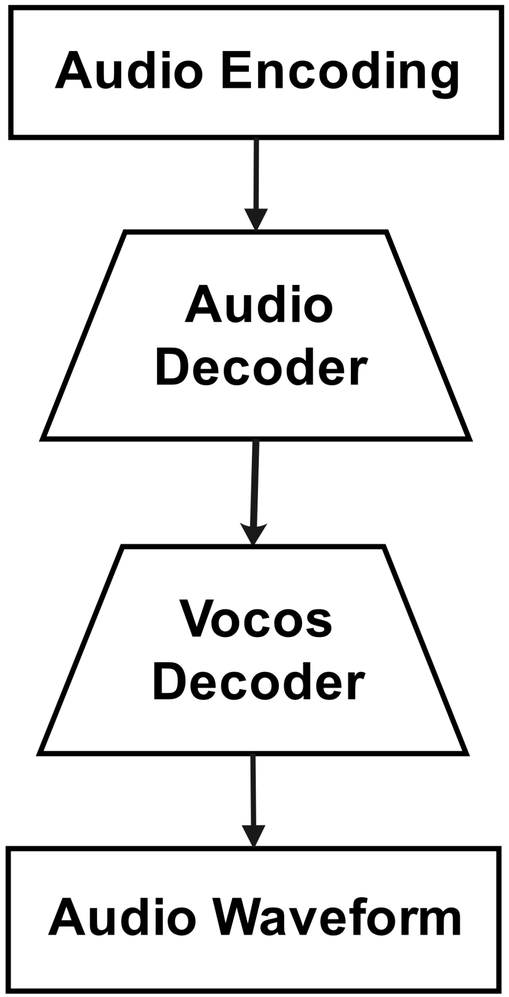
Audio Autoencoder - Demo - Latent Mixing

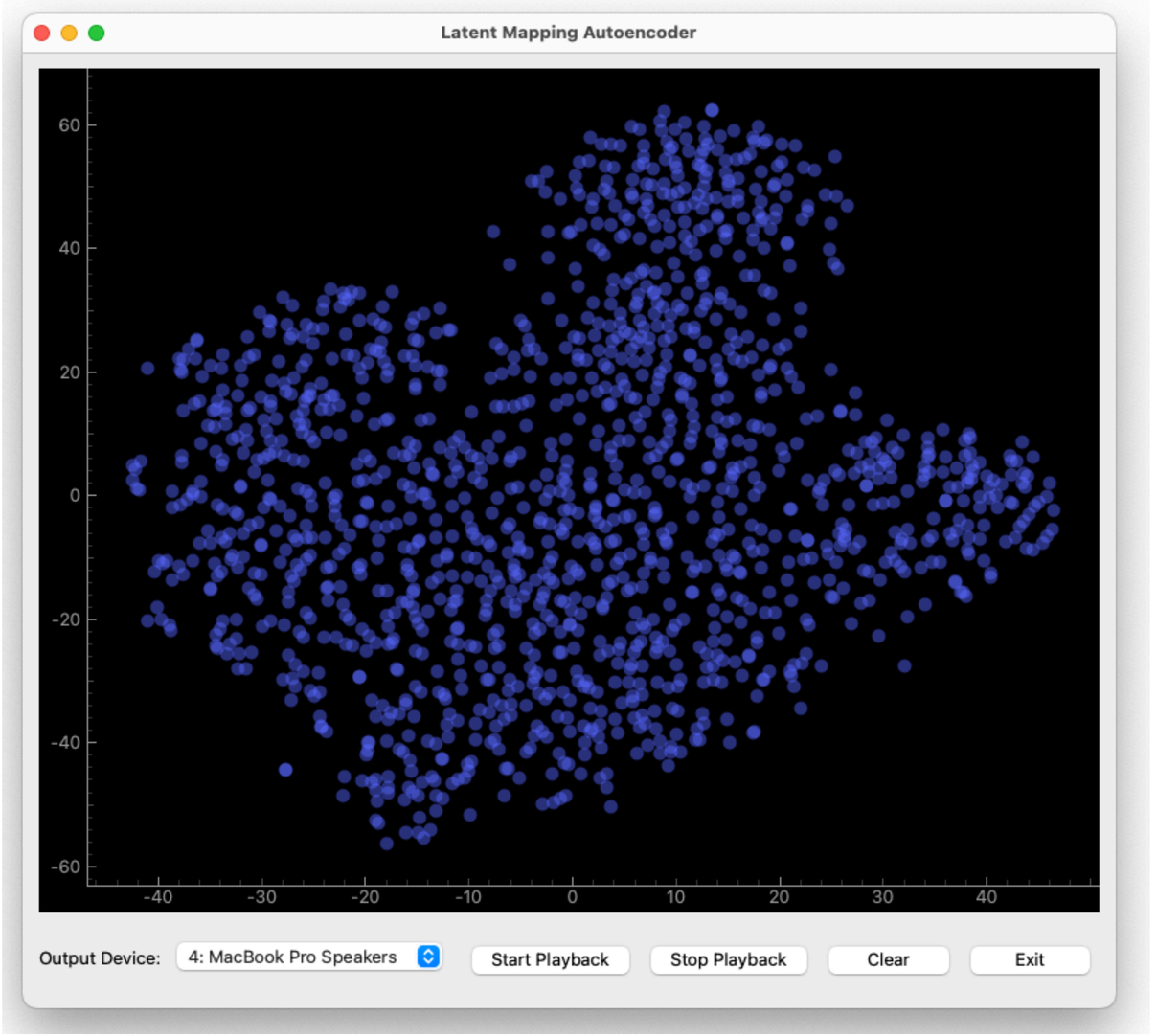




AI_Sound / Autoencoder / Inference / LatentManipulation / vae_vocos_cnn_interactive.py

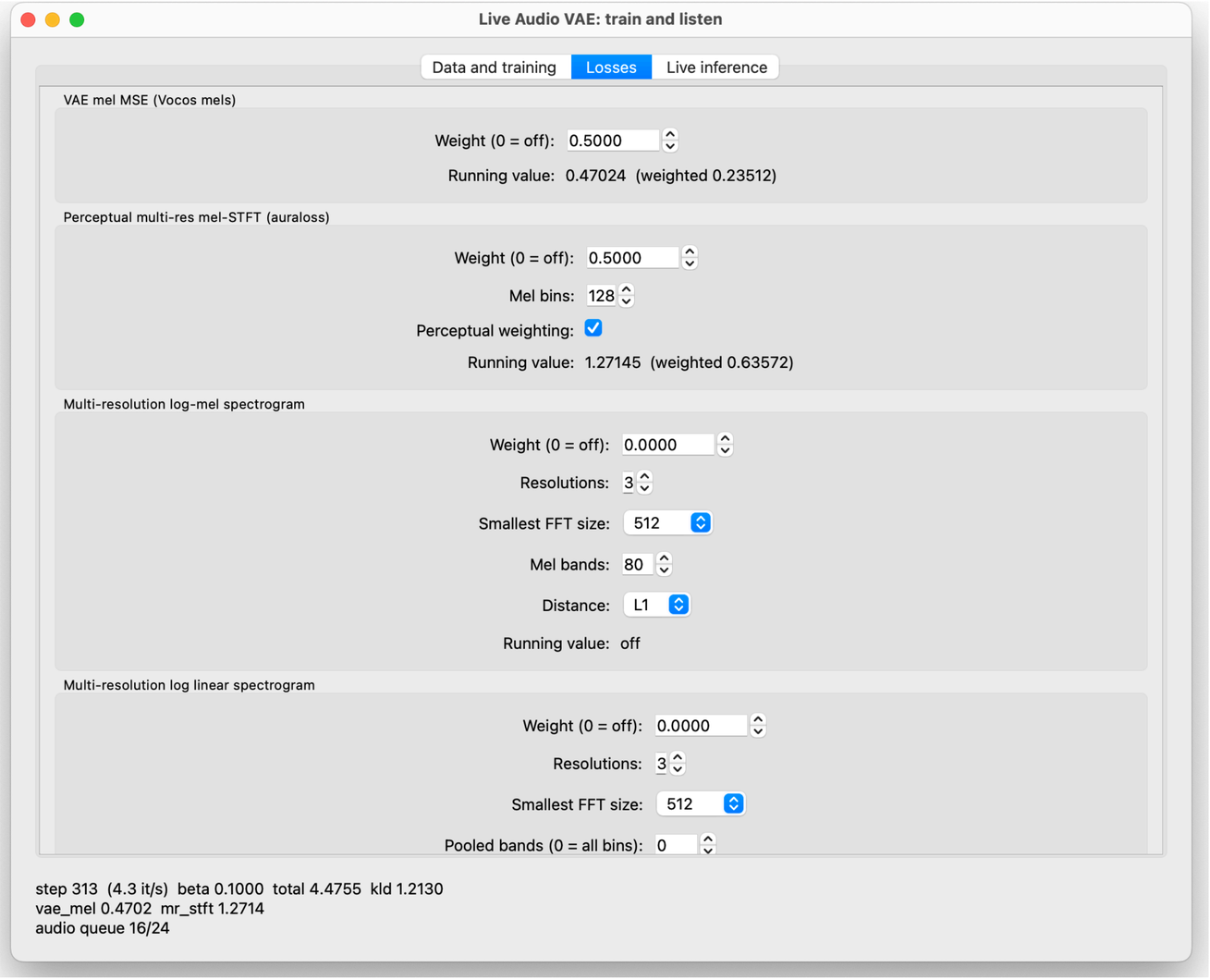
Audio Autoencoder - Demo - Latent Navigation





AI_Sound / Autoencoder / Inference / LatentNavigation / vae_vocos_cnn_interactive.py

Audio Autoencoder - Demo - Realtime Training



AI_Sound / Autoencoder / Training /
audio_vae_vocos_cnn_live_train.py

Audio Autoencoder V2

- Larger Audio Window
- Preserve Temporality in Latent Representation
- Deeper Architecture with Residual Connections
- Adversarial Training
- Vocos finetuning